



2807 West County Road 75
Monticello, MN 55362

September 24, 2025

L-MT-25-034
10 CFR 50.73

ATTN: Document Control Desk
U.S. Nuclear Regulatory Commission
Washington, DC 20555-0001

Monticello Nuclear Generating Plant
Docket No. 50-263
Subsequent Renewed Facility Operating License No. DPR-22

Monticello Nuclear Generating Plant (MNGP) Licensee Event Report 2025-001-01

Northern States Power Company, a Minnesota corporation, doing business as Xcel Energy (hereafter "NSPM"), hereby submits a supplement to MNGP Licensee Event Report (LER) 2025-001-00 in accordance with 10 CFR 50.73(a)(2)(iv)(A).

If you have any questions about this submittal, please contact Rick Loeffler, Senior Nuclear Regulatory Engineer, by email at rick.a.loeffler@xcelenergy.com, or by phone at 612-539-3370.

Summary of Commitments

This letter makes no new commitments and no revisions to existing commitments.

A handwritten signature in black ink, appearing to read 'Michelle Neal'.

Michelle Neal
Plant Manager, Monticello Nuclear Generating Plant
Northern States Power Company – Minnesota

Enclosure

cc: Administrator, Region III, USNRC
Project Manager, Monticello, USNRC
Resident Inspector, Monticello, USNRC
State of Minnesota

ENCLOSURE

**MONTICELLO NUCLEAR GENERATING PLANT
LICENSEE EVENT REPORT 2025-001-01**

5 pages follow



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form)

<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Library, and Information Collections Branch (T-6 A10M), U. S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by email to Infocollections.Resource@nrc.gov, and the OMB reviewer at: OMB Office of Information and Regulatory Affairs, (3150-0104), Attn: Desk Officer for the Nuclear Regulatory Commission, 725 17th Street NW, Washington, DC 20503. The NRC may not conduct or sponsor, and a person is not required to respond to, a collection of information unless the document requesting or requiring the collection displays a currently valid OMB control number.

1. Facility Name Monticello Nuclear Generating Plant	☒ 050 ☐ 052	2. Docket Number 263	3. Page 1 OF 5
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4. Title
Automatic Reactor Scram on Low Water Level

5. Event Date			6. LER Number			7. Report Date			8. Other Facilities Involved		
Month	Day	Year	Year	Sequential Number	Revision No.	Month	Day	Year	Facility Name	☐ 050	Docket Number
03	19	2025	2025	- 001 -	01	09	24	2025	Facility Name	☐ 052	Docket Number

9. Operating Mode 1	10. Power Level 97 percent
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11. This Report is Submitted Pursuant to the Requirements of 10 CFR §: (Check all that apply)

☒ 10 CFR Part 20	☐ 20.2203(a)(2)(vi)	☒ 10 CFR Part 50	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)	☐ 73.1200(a)
☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)	☐ 73.1200(b)
☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)	☐ 73.1200(c)
☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.36(c)(2)	☒ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)	☐ 73.1200(d)
☐ 20.2203(a)(2)(i)	☒ 10 CFR Part 21	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(A)	☒ 10 CFR Part 73	☐ 73.1200(e)
☐ 20.2203(a)(2)(ii)	☐ 21.2(c)	☐ 50.69(g)	☐ 50.73(a)(2)(v)(B)	☐ 73.77(a)(1)	☐ 73.1200(f)
☐ 20.2203(a)(2)(iii)		☐ 50.73(a)(2)(i)(A)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(2)(i)	☐ 73.1200(g)
☐ 20.2203(a)(2)(iv)		☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(ii)	☐ 73.1200(h)
☐ 20.2203(a)(2)(v)		☐ 50.73(a)(2)(i)(C)	☐ 50.73(a)(2)(vii)		
☐ OTHER (Specify here, in abstract, or NRC 366A).					

12. Licensee Contact for this LER

Licensee Contact Richard A. Loeffler, Senior Nuclear Regulatory Engineer	Phone Number (Include area code) 612-539-3370
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13. Complete One Line for each Component Failure Described in this Report

Cause	System	Component	Manufacturer	Reportable to IRIS	Cause	System	Component	Manufacturer	Reportable to IRIS
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

14. Supplemental Report Expected

☒ No	☐ Yes (If yes, complete 15. Expected Submission Date)	15. Expected Submission Date	Month	Day	Year
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16. Abstract (Limit to 1326 spaces, i.e., approximately 13 single-spaced typewritten lines)

On March 19, 2025, at 1136 Central Daylight Time (CDT), an automatic reactor scram occurred due to low reactor pressure vessel (RPV) water level due to a trip of the 11 Reactor Feedwater Pump (RFP). This event is reportable pursuant to 10 CFR 50.73(a)(2)(iv)(A) as an automatic actuation of the Reactor Protection System and due to primary containment isolation valves closure on a Group 2 isolation signal. All nuclear safety systems functioned as designed.

The cause of the scram was low RPV water level due to the trip of the 11 RFP. The 11 RFP tripped on low suction pressure due to low water level within the Condenser Hotwell and air binding in the Circulating Water System (CWS). The Condenser tube sheets were cleaned, the eight vacuum air release valves (VARs) were replaced, and the Hotwell Makeup Level Controller was replaced.

Corrective actions to minimize recurrence include: replacing the VARs each refueling outage, cleaning, as required, the main Condenser each fall and spring, recording and trending silt accumulation, and strengthening procedural guidance for identification and mitigation of reduced Condenser and CWS margin.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
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1. FACILITY NAME Monticello Nuclear Generating Plant	☒ 050	2. DOCKET NUMBER 263	3. LER NUMBER		
	☐ 052		YEAR 2025	SEQUENTIAL NUMBER 001	REV NO. 01

NARRATIVE**I. Plant Operating Conditions before the Event**

On March 19, 2025, the unit was operating in Mode 1 at 97 of percent of Rated Thermal Power (RTP).

II. Description of the Event**A. Event Summary**

On March 19, 2025, at 1136 Central Daylight Time (CDT), an automatic reactor scram occurred due to low reactor pressure vessel (RPV) water level due to a trip of the 11 Reactor Feedwater Pump [EIS Code: SJ, P]. The scram was uncomplicated with all systems responding as expected.

Valid actuations of the Reactor Protection System (RPS) [JC] and the Primary Containment Isolation System (PCIS) [JM] occurred as the Containment isolation valves actuated and closed on a valid Group 2 isolation signal.

This event is reportable pursuant to 10 CFR 50.73(a)(2)(iv)(A) as an automatic actuation of the RPS and the PCIS.

B. Status of the structures, systems, or components (SSCs) that were inoperable at the start of the event and that contributed to the event.

There were no SSCs whose inoperability contributed to this event.

C. Dates and approximate times of occurrences

March 19, 2025, at 1136 CDT, an automatic reactor scram on low RPV water level occurred due to a Feedwater Pump trip. (EN# 57618)

D. Manufacturer and model number of each component that failed during the event.

N/A

E. Other systems or secondary functions affected

No other systems or secondary functions were affected.

F. Method of discovery of each component or system failure or procedural error.

Degrading Condenser [SG] performance was present during the course of the event.

Stuck/degraded vacuum air release valves [RV] were identified during Condenser cleaning and inspection.

Previously identified Hotwell level controller performance issues were identified during the post-scram review.

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1. FACILITY NAME

Monticello Nuclear Generating Plant



050



052

2. DOCKET NUMBER

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01

NARRATIVE**G. The failure mode, mechanism, and effect of each failed component.**

The 11 RFP tripped on low suction pressure due to low water level within the Condenser Hotwell.

Stuck/degraded (partially filled) vacuum air release valves on the Condenser reduced the scavenging system capability to remove air from the inner loop water boxes.

Condenser Hotwell level control was challenged due to issues with the level controller.

H. Operator actions

Operations personnel stabilized the plant in Mode 3 following the reactor scram.

I. Automatically and manually initiated safety system responses.

RPS actuation and PCIS Group 2 isolation signals received, and the components actuated as required.

III. Cause of the Event**A. Cause of each component or system failure or personnel error.**

The cause of the scram was low RPV water level due to a trip of the 11 RFP. The 11 RFP tripped on low suction pressure due to low water level within the Condenser Hotwell and air binding in the Circulating Water System.

Several different issues resulted in air binding of the Condenser. Condenser tube sheet fouling caused air binding to occur in the condenser outlet piping. Stuck/degraded vacuum air release valves reduced the scavenging system capability to remove air from the inner loop water boxes. Hotwell level control issues challenged and increased the impact of the air binding within the Condenser and Circulating Water System.

B. Cause(s) and circumstances for each human performance related root cause.

No human performance errors were identified during the investigation.

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Monticello Nuclear Generating Plant		263	YEAR	SEQUENTIAL NUMBER	REV NO.
			2025	001	01

NARRATIVE**IV. Analysis of the Event**

On March 18, 2025, at approximately 18:18 CDT, the low pressure (LP) and high pressure (HP) Condenser began exhibiting degraded performance. At approximately 18:45 CDT, Operations entered the procedures for decreased circulating water flow and degrading condenser vacuum, due to the condenser performance changes. Attempts to improve Condenser performance in accordance with plant procedures were unsuccessful. On March 19, 2025, at approximately 11:35 CDT, the 11 RFP tripped. At 11:36 CDT the reactor scrambled when the low RPV water level setpoint was reached.

Subsequently, during the Condenser inspection and cleanup it was determined that several vacuum air release valves were partially filled with debris and Amertap balls. As part of the event investigation, it was identified that a Hotwell Makeup Level Controller did not perform as expected.

V. Assessment of Safety Consequences

The plant responded as designed, while maintaining defense-in-depth for nuclear safety. All nuclear safety systems functioned as designed. This event was of very low nuclear safety significance. At no time was the health and safety of the public at risk.

A. Availability of systems or components that could have performed the same function as the components and systems that failed during the event.

N/A

B. For events that occurred when the reactor was shut down, the availability of systems or components needed to shutdown the reactor and maintain safe shutdown conditions, remove residual heat, control the release of radioactive material, or mitigate the consequences of an accident.

This event did not occur when the reactor was shut down.

C. For failure that rendered a train of a safety system inoperable, estimate of the elapsed time from discovery of the failure until the train was returned to service.

There were no safety systems rendered inoperable.

VI. Corrective Actions**A. Immediate Corrective Actions**

- 1) The Condenser was cleaned.
- 2) All eight vacuum air release valves were replaced.
- 3) The Hotwell Makeup Level Controller was replaced.

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NARRATIVE

Following the repairs and cleaning of the Condenser the reactor was restarted and synced to the electrical grid on March 22, 2025.

B. Corrective Actions to Prevent Recurrence or to reduce the probability of similar events occurring in the future.

The following procedural and/or scheduling activities have been put in place:

- 1) Vacuum air release valves are scheduled to be replaced each refueling outage.
- 2) Clean Condenser each fall and spring, as required, based on evaluation of plant parameters.
- 3) Procedural changes to modify dredging activities to record and trend silt accumulations and clarify the acceptance criteria for dredging.
- 4) Procedural changes to strengthen the guidance for identification and mitigation of reduced Condenser and CWS margin.

VII. Previous Similar Events at the Same Site

A search of LERs for the MNGP identified no previous similar events in the past three years.

VIII. Additional Information

There is no additional information.

Energy Industry Identification System (EIIIS) codes are identified in the text as [XX].